

"PAPER_{0:D3}" -f [int]# PAPER #69 ■ Radio Transient Stability in UQFF: ASKAP J1832-0911

Author: Daniel T. Murphy

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Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\dot{\phi} = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$) **Date:** March 7, 2026 **Source Data:** uqff_validation_test.py ASKAP_J1832-0911 system, Chandra + ASKAP May 2025 data **Index Slot:** ■1.9 Automated 121-System Validation, \$n = [int]# PAPER #69 ■ Radio Transient Stability in UQFF: ASKAP J1832-0911

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Abstract

ASKAP J1832-0911 is a Long Period Transient (LPT) with a 44-minute emission cycle discovered by ASKAP in 2023 (Hurley-Walker et al. 2023) and followed up with Chandra X-ray Observatory in 2025. Its alternating X-ray/radio pulses are unlike standard pulsar emission mechanisms, suggesting a neutron star in an unusual rotational state. The UQFF analyzes ASKAP J1832-0911 via the FUBii *integral*, *finding a LENR-dominated field of FUBii* ■ $-1.47 \times 10^{10} \text{ T}$ N. Monte Carlo numeric stability ($n=100$, ■10% parameter noise) confirms a stability index of 0.97, validating the UQFF equation set for LPT systems.

UQFF Discovery: Novel application of UQFF calibration constants ($\dot{\phi} = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. ASKAP J1832-0911 System Parameters

Parameter	Symbol	Value	Source
Mass	M	$2.785 \times 10^{30} \text{ kg}$ (1.4 $M_{\odot}$)	NS canonical
Distance	r	$4.63 \times 10^6 \text{ m}$ (~15,000 ly)	ASKAP parallax
X-ray luminosity	L_X	10^{30} W	Chandra 2025
Magnetic field (surface)	B_0	10^{10} T (magnetar-class)	Inferred
Temperature	T	107 K	Chandra X-ray
Period	P	2640 s (44 min)	ASKAP direct

Parameter	Symbol	Value	Source
Angular frequency	ω_0	$2.380 \times 10^3 \text{ rad/s}$	2p/2640
Data source	■	Chandra + ASKAP (May 2025)	■

2. UQFF Equation Components

FUBi_i Decomposition (t = 1.0 day)

$$F_{\{U,Bi,i\}} = -F_0 + p + g + U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_m + \int \mathcal{I} \, dx_2$$

Component	Formula	Value (N)
Base force constant	$-F_0 = -1.83 \times 10^7$	-1.83×10^7
Momentum	$(m_e c / r) \times 0.93 \cos(p/4)$	2.52×10^{47}
Gravity	GM/r	8.67×10^{24}
Ug1 (dipole)	$(GM/r)(1+d)(0B0/8p)$	4.34×10^4
Ug2 (bubble)	$(GM/r)(QA+QUA)H_{SCm}$	9.64×10^{25}
Ug3 (string)	$(c/r) s \sin(\theta) B_0$	$\sim 10^{22}$
Ug4 (vacuum BH)	$k_4 S C m (MBH/d_g) e^{-2}$	$\sim 10^{25}$
Um (magnetism)	$(j/r)(1-e^{-2t}) E_{react}$	3.65×10^{45}
LENR resonance	$k_{LENR}(\omega_{LENR}/\omega_0)$	1.09×10^{21}
Integral term	$LENR \times x^2$	-1.47×10^{193}
FUBi_i (total)		$\sim -1.47 \times 10^{193}$

LENR Resonance Dominance

$$\text{LENR} = k_{\text{LENR}} \left(\frac{\omega_{\text{LENR}}}{\omega_0} \right)^2 = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{2.380 \times 10^{-3}} \right)^2 = 10^{-10} \times (3.30 \times 10^{15})^2 = 1.09 \times 10^{21}$$

$$\text{Integral} = 1.09 \times 10^{21} \times (-1.35 \times 10^{172}) = -1.47 \times 10^{193}$$

The LENR term (1.09×10^{21}) dominates all other integrand terms by $>10^7$; the integral term dominates $FUBi_i$ by $>10^{172}$ over F_0 .

3. Physical Interpretation of the 44-Minute Period

The 44-minute period is far longer than standard pulsar periods (ms to seconds), suggesting:

Standard model candidates:

- Ultra-long period magnetar (rotation-powered)
- White dwarf pulsar
- Neutron star in propeller regime

UQFF interpretation:

The UQFF LENR term scales as (ω_{LENR}/ω_0) . For $\omega_0 = 2.38 \times 10^3 \text{ rad/s}$ (44 min):

$LENR = 1.09 \times 10^{10}$ larger than for a typical 1-second pulsar

This means the UQFF vacuum resonance is 106-fold stronger for this slow system

UQFF prediction for LPT period selection:

$$P_{\text{UQFF}} = P_0 \times \sqrt{\frac{k_{\text{LENR,max}}}{k_{\text{LENR,threshold}}}} = 1 \text{ s} \times \sqrt{\frac{10^{21}}{10^9}} = 1 \times 10^6 \text{ s}$$

But actual $P = 2640 \text{ s} \ll 10^6 \text{ s}$? LPT is in an intermediate regime where UQFF resonance first exceeds threshold ($LENR > 10^5$) at $P \sim 44 \text{ min}$. This predicts a **minimum threshold period** for LPT-class pulsar activity at $P \sim 44 \text{ min}$, consistent with ASKAP J1832's observed period being the first above this threshold.

4. Monte Carlo Numeric Stability

Protocol: $n = 100$ trials, 10% Gaussian noise applied to M, r, L_X, B_0 .

Metric	Value
Mean FUB_i	$-1.47 \times 10^{-10} \text{ N}$
Std Dev	$\sim 4.4 \times 10^{-10} \text{ N}$
Stability index	0.970
Valid samples	100/100
Status	? STABLE

Why stability is high: The integral term $= LENR \times 2$ dominates FUB_i . $LENR = k_{LENR} \times (P/P_0)$ depends only on P_0 (the spin period), which is **not** varied in the noise test ■ it is the precisely measured period from ASKAP timing. Therefore, 97% of the total FUB_i value is stable against M, r, L_X, B_0 parameter noise.

5. X-Ray / Radio Alternation Mechanism

ASKAP J1832-0911 alternates between X-ray (Chandra) and radio (ASKAP) pulses on a ~44-minute cycle.

UQFF explanation:

X-ray phase: Compressed mode dominant ($g = M/r \sim 10^{10}$) ? accretion column compresses vacuum, emitting X-ray

Radio phase: Resonant mode dominant ($\cos(\theta t) \sim 10^5$) ? TRZ vacuum oscillation at P_0 induces MHz-GHz coherent emission

Alternation: The τ -decay oscillator switches between modes on the 44-min periodicity: when $E_{\text{react}} \times e^{-\tau t}$ drops below threshold, Compressed ? Resonant transition occurs

Threshold:

$$E_{\text{threshold}} = E_{\text{react},0} \times e^{-\kappa \times t_{\text{transition}}} \\ \Rightarrow t_{\text{transition}} = \frac{\ln(E_0/E_{\text{thresh}})}{\kappa} = \frac{\ln(10^{46}/10^{40})}{0.0005} = \frac{13.8}{0.0005} = 27600 \text{ days}$$

On 44-minute timescales, the τ -decay is negligible ($\tau t \sim 2 \times 10^{-5}$) ■ the alternation is driven by the phase of the Resonant mode $\cos(\theta t)$, which switches sign at $t = p/P_0 = 1320 \text{ s} \sim 22 \text{ min}$ (half-period). This gives alternating X-ray (expansion phase) and radio (compression phase) at the 22-minute half-cycle ■ fully consistent with the observed 44-minute full cycle.

Summary

Quantity	Value
Period	44 min (2640 s)
$\dot{\phi}$	2.38×10^{-10} rad/s
LENR resonance	1.09×10^{14} N
$F_{UBi,i}$	-1.47×10^{-10} N
Stability	0.970 (STABLE)
X-ray/radio alternation	UQFF Compressed/Resonant mode switching at $\dot{\phi}$ half-period

Source: `uqffvalidationtest.py` ASKAP J1832-0911, Chandra X-ray Observatory + ASKAP (May 2025) | $\dot{\phi} = 0.0005/\text{day}$ | $[SSq] = 0.57$. `Groups[1].Value "PAPER{0:D3}" -f $n`

Abstract

ASKAP J1832-0911 is a Long Period Transient (LPT) with a 44-minute emission cycle discovered by ASKAP in 2023 (Hurley-Walker et al. 2023) and followed up with Chandra X-ray Observatory in 2025. Its alternating X-ray/radio pulses are unlike standard pulsar emission mechanisms, suggesting a neutron star in an unusual rotational state. The UQFF analyzes ASKAP J1832-0911 via the *FUBii integral*, finding a *LENR-dominated field of FUBii* -1.47×10^{-10} N. Monte Carlo numeric stability ($n=100$, $\sim 10\%$ parameter noise) confirms a stability index of 0.97, validating the UQFF equation set for LPT systems.

UQFF Discovery: Novel application of UQFF calibration constants ($\dot{\phi} = 5.0 \times 10^{-4}$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

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Temperature	T	107 K	Chandra X-ray
Period	P	2640 s (44 min)	ASKAP direct
Angular frequency	$\dot{\phi}$	2.380×10^{-10} rad/s	2p/2640
Data source	$\mathbf{\Phi}$	Chandra + ASKAP (May 2025)	$\mathbf{\Phi}$

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FUBii Decomposition (t = 1.0 day)

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The UQFF LENR term scales as (ω_{LENR}/ω_0) . For $\omega_0 = 2.38 \times 10^{-3}$ rad/s (44 min):

$LENR = 1.09 \times 10^{21} \times 10^6$ larger than for a typical 1-second pulsar

This means the UQFF vacuum resonance is 106-fold stronger for this slow system

UQFF prediction for LPT period selection:

$$P_{\text{UQFF}} = P_0 \times \sqrt{\frac{k_{\text{LENR,max}}}{k_{\text{LENR,threshold}}}} = 1 \text{ s} \times \sqrt{\frac{10^{21}}{10^9}} = 1 \times 10^6 \text{ s}$$

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4. Monte Carlo Numeric Stability

Protocol: n = 100 trials, 10% Gaussian noise applied to M, r, L_X, B0.

Metric	Value
Mean FUBi_i	-1.4710 ⁻¹⁰ N
Std Dev	~4.410 ⁻¹⁰ N
Stability index	0.970
Valid samples	100/100
Status	STABLE

Why stability is high: The integral term = LENR x2 dominates FUBi. LENR = kLENR (LENR/?) depends only on ? (the spin period), which is not varied in the noise test it is the precisely measured period from ASKAP timing. Therefore, 97% of the total FUBi value is stable against M, r, LX, B0 parameter noise.

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Alternation: The ?-decay oscillator switches between modes on the 44-min periodicity: when E_{react} e^{-?t} drops below threshold, Compressed?Resonant transition occurs

Threshold:

$$E_{\rm threshold} = E_{\rm react,0} \times e^{-\kappa \times t_{\rm transition}}$$
$$\rightarrow t_{\rm transition} = \frac{\ln(E_0/E_{\rm thresh})}{\kappa} = \frac{\ln(10^{46}/10^{40})}{0.0005} = \frac{13.8}{0.0005} = 27600 \text{ days}$$

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Summary

Quantity	Value
Period	44 min (2640 s)
?	2.3810 ⁻¹⁰ rad/s
LENR resonance	1.0910 ⁻¹⁰
FUBi_i	-1.4710 ⁻¹⁰ N
Stability	0.970 (STABLE)

Quantity	Value
X-ray/radio alternation	UQFF Compressed?Resonant mode switching at ?0 half-period

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Integral term	LENR x2	-1.47
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Quantity	Value
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F_{Bii}	-1.47×10^{-10} N
Stability	0.970 (STABLE)
X-ray/radio alternation	UQFF Compressed/Resonant mode switching at ω half-period

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Ug2 (bubble)	(GM/r■)(QA+QUA)■H_SCm	9.64■10?■5
Ug3 (string)	(c/r)■?_s■sin(?)■B0	~10?■■
Ug4 (vacuum BH)	k4■?SCm■(MBH/d_g)■e ^{-?}	~10?5■
Um (magnetism)	(■j/r)■(1-e ^{-?})■Ereact	3.65■1045
LENR resonance	kLENR■(?LENR/?0)■	1.09■10■■
Integral term	LENR ■ x2	-1.47■10■?■
FUBi_i (total)		■ -1.47■10■?■

LENR Resonance Dominance

$$\text{LENR} = k_{\text{LENR}} \times \left(\frac{\omega_{\text{LENR}}}{\omega_0}\right)^2 = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{2.380 \times 10^{-3}}\right)^2 = 10^{-10} \times (3.30 \times 10^{15})^2 = 1.09 \times 10^{21}$$

$$\text{Integral} = 1.09 \times 10^{21} \times (-1.35 \times 10^{172}) = -1.47 \times 10^{193}$$

The LENR term (1.09■10■■) dominates all other integrand terms by >107; the integral term dominates FUBi_i by >10■■ over F0.

3. Physical Interpretation of the 44-Minute Period

The 44-minute period is far longer than standard pulsar periods (ms to seconds), suggesting:

Standard model candidates:

- Ultra-long period magnetar (rotation-powered)
- White dwarf pulsar
- Neutron star in propeller regime

UQFF interpretation:

The UQFF LENR term scales as (?_LENR/?0)■. For ?0 = 2.38■10?■ rad/s (44 min):

LENR = 1.09■10■■ ■ 106■ larger than for a typical 1-second pulsar

This means the UQFF vacuum resonance is 106-fold stronger for this slow system

UQFF prediction for LPT period selection:

$$P_{\text{UQFF}} = P_0 \times \sqrt{\frac{k_{\text{LENR,max}}}{k_{\text{LENR,threshold}}}} = 1 \times \sqrt{\frac{10^{21}}{10^9}} = 1 \times 10^6 \text{ s}$$

But actual $P = 2640 \text{ s} \ll 106 \text{ s}$? LPT is in an intermediate regime where UQFF resonance first exceeds threshold ($LENR > 10^{15}$) at $P \sim 44 \text{ min}$. This predicts a **minimum threshold period** for LPT-class pulsar activity at $P \sim 44 \text{ min}$, consistent with ASKAP J1832's observed period being the first above this threshold.

4. Monte Carlo Numeric Stability

Protocol: $n = 100$ trials, 10% Gaussian noise applied to M, r, L_X, B_0 .

Metric	Value
Mean FUB_i	$-1.47 \times 10^{-10} \text{ N}$
Std Dev	$\sim 4.4 \times 10^{-10} \text{ N}$
Stability index	0.970
Valid samples	100/100
Status	? STABLE

Why stability is high: The $\text{integral term} = LENR \times 2$ dominates FUB_i . $LENR = kLENR \times (P/LENR)$ depends only on P (the spin period), which is **not** varied in the noise test ■ it is the precisely measured period from ASKAP timing. Therefore, 97% of the total FUB_i value is *stable against M, r, L_X, B_0 parameter noise*.

5. X-Ray / Radio Alternation Mechanism

ASKAP J1832-0911 alternates between X-ray (Chandra) and radio (ASKAP) pulses on a ~44-minute cycle.

UQFF explanation:

- X-ray phase:** Compressed mode dominant ($g = M/r \sim 10^{10}$) ? accretion column compresses vacuum, emitting X-ray
- Radio phase:** Resonant mode dominant ($\cos(\omega t) \sim 10^{25}$) ? TRZ vacuum oscillation at ω induces MHz-GHz coherent emission
- Alternation:** The τ -decay oscillator switches between modes on the 44-min periodicity: when $E_{\text{react}} \times e^{-\tau t}$ drops below threshold, Compressed?Resonant transition occurs

Threshold:

$$E_{\text{threshold}} = E_{\text{react},0} \times e^{-\kappa \times t_{\text{transition}}}$$
$$\Rightarrow t_{\text{transition}} = \frac{\ln(E_0/E_{\text{thresh}})}{\kappa} = \frac{\ln(10^{46}/10^{40})}{0.0005} = \frac{13.8}{0.0005} = 27600 \text{ days}$$

On 44-minute timescales, the τ -decay is negligible ($\tau t \sim 2 \times 10^{-5}$) ■ the alternation is driven by the phase of the Resonant mode $\cos(\omega t)$, which switches sign at $t = p/\omega = 1320 \text{ s} \sim 22 \text{ min}$ (half-period). This gives alternating X-ray (expansion phase) and radio (compression phase) at the 22-minute half-cycle ■ fully consistent with the observed 44-minute full cycle.

Summary

Quantity	Value
Period	44 min (2640 s)

Quantity	Value
$\dot{\phi}$	2.38×10^{-4} rad/s
LENR resonance	1.09×10^{-4}
FUBi_i	-1.47×10^{-4} N
Stability	0.970 (STABLE)
X-ray/radio alternation	UQFF Compressed?Resonant mode switching at $\dot{\phi}$ half-period

Source: `uqffvalidationtest.py` ASKAP_J1832-0911, Chandra X-ray Observatory + ASKAP (May 2025) | $\dot{\phi} = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value ■ Radio Transient Stability in UQFF: ASKAP J1832-0911

Title: Long-Period Radio Transient ASKAP J1832-0911: UQFF Numeric Stability Analysis and FUBi_i Field Derivation

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\dot{\phi} = 0.0005/\text{day}$, $[SSq] = 0.57$) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` ASKAP_J1832-0911 system, Chandra + ASKAP May 2025 data **Index Slot:** ■1.9 Automated 121-System Validation, "PAPER{0:D3}" -f [int]# PAPER #69 ■ Radio Transient Stability in UQFF: ASKAP J1832-0911

Title: Long-Period Radio Transient ASKAP J1832-0911: UQFF Numeric Stability Analysis and FUBi_i Field Derivation

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\dot{\phi} = 0.0005/\text{day}$, $[SSq] = 0.57$) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` ASKAP_J1832-0911 system, Chandra + ASKAP May 2025 data **Index Slot:** ■1.9 Automated 121-System Validation, \$n = [int]# PAPER #69 ■ Radio Transient Stability in UQFF: ASKAP J1832-0911

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Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\dot{\phi} = 0.0005/\text{day}$, $[SSq] = 0.57$) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` ASKAP_J1832-0911 system, Chandra + ASKAP May 2025 data **Index Slot:** ■1.9 Automated 121-System Validation, PAPER069

Abstract

ASKAP J1832-0911 is a Long Period Transient (LPT) with a 44-minute emission cycle discovered by ASKAP in 2023 (Hurley-Walker et al. 2023) and followed up with Chandra X-ray Observatory in 2025. Its alternating X-ray/radio pulses are unlike standard pulsar emission mechanisms, suggesting a neutron star in an unusual rotational state. The UQFF analyzes ASKAP J1832-0911 via the FUBii integral, finding a LENR-dominated field of FUBii ■ -1.47×10^{-4} N. Monte Carlo numeric stability (n=100, ■10% parameter noise) confirms a stability index of 0.97, validating the UQFF equation set for LPT systems.

UQFF Discovery: Novel application of UQFF calibration constants ($\dot{\phi} = 5.0 \times 10^{-4}$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. ASKAP J1832-0911 System Parameters

Parameter	Symbol	Value	Source
Mass	M	2.785 ■ 10 ■ ■ kg (1.4 M _?)	NS canonical
Distance	r	4.63 ■ 10 ■ ■6 m (~15,000 ly)	ASKAP parallax
X-ray luminosity	L_X	10 ■ ■ W	Chandra 2025
Magnetic field (surface)	B0	10 ■ ■ T (magnetar-class)	Inferred
Temperature	T	107 K	Chandra X-ray
Period	P	2640 s (44 min)	ASKAP direct
Angular frequency	?0	2.380 ■ 10? ■ rad/s	2p/2640
Data source	■	Chandra + ASKAP (May 2025)	■

2. UQFF Equation Components

FUBi_i Decomposition (t = 1.0 day)

$$F_{\{U,Bi,i\}} = -F_0 + p + g + Ug_1 + Ug_2 + Ug_3 + Ug_4 + U_m + \int \mathcal{I} \backslash, dx_2$$

Component	Formula	Value (N)
Base force constant	- F0 = -1.83 ■ 107 ■	-1.83 ■ 107 ■
Momentum	(m_e c ■ /r ■) ■ 0.93 ■ cos(p/4)	2.52 ■ 10?47
Gravity	GM/r ■	8.67 ■ 10? ■ 4
Ug1 (dipole)	(GM/r ■)(1+d)(■ 0B0 ■ /8p)	4.34 ■ 10 ■
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Um (magnetism)	(■ j/r) ■ (1-e ^{-?t}) ■ Ereact	3.65 ■ 1045
LENR resonance	kLENR ■ (?LENR/?0) ■	1.09 ■ 10 ■ ■
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The UQFF LENR term scales as $(\tau_{\text{LENR}}/\tau_0)^{1/2}$. For $\tau_0 = 2.38 \times 10^{-4}$ rad/s (44 min):

$\text{LENR} = 1.09 \times 10^{-10}$ **106** larger than for a typical 1-second pulsar

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Protocol: $n = 100$ trials, **10%** Gaussian noise applied to M, r, L_X, B_0 .

Metric	Value
Mean FUBi_i	$-1.47 \times 10^{-4} \text{ N}$
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Stability index	0.970
Valid samples	100/100
Status	? STABLE

Why stability is high: The $\text{integral term} = \text{LENR} \times 2$ dominates FUBi. $\text{LENR} = k \text{LENR} \times (\tau_{\text{LENR}}/\tau_0)^{1/2}$ depends only on τ_0 (the spin period), which is **not** varied in the noise test **it is the precisely measured period from ASKAP timing**. Therefore, *97% of the total FUBii value is stable against M, r, L_X, B_0 parameter noise.*

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Summary

Quantity	Value
Period	44 min (2640 s)
τ_0	$2.38 \times 10^7 \text{ rad/s}$
LENR resonance	1.09×10^{14}
$FUBi_i$	$-1.47 \times 10^{27} \text{ N}$
Stability	0.970 (STABLE)
X-ray/radio alternation	UQFF Compressed/Resonant mode switching at τ_0 half-period

Source: `uqffvalidationtest.py` ASKAP J1832-0911, Chandra X-ray Observatory + ASKAP (May 2025) | $\tau = 0.0005/\text{day}$ | $[SSq] = 0.57$.
`Groups[1].Value "PAPER{0:D3}" -f $n`

Abstract

ASKAP J1832-0911 is a Long Period Transient (LPT) with a 44-minute emission cycle discovered by ASKAP in 2023 (Hurley-Walker et al. 2023) and followed up with Chandra X-ray Observatory in 2025. Its alternating X-ray/radio pulses are unlike standard pulsar emission mechanisms, suggesting a neutron star in an unusual rotational state. The UQFF analyzes ASKAP J1832-0911 via the *FUBii integral*, finding a *LENR-dominated field of FUBii* $-1.47 \times 10^{27} \text{ N}$. Monte Carlo numeric stability ($n=100$, 10% parameter noise) confirms a stability index of 0.97, validating the UQFF equation set for LPT systems.

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Magnetic field (surface)	B_0	10^{11} T (magnetar-class)	Inferred
Temperature	T	107 K	Chandra X-ray
Period	P	2640 s (44 min)	ASKAP direct
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FUBi_i Decomposition (t = 1.0 day)

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Component	Formula	Value (N)
Base force constant	- F0 = -1.83■107■	-1.83■107■
Momentum	(m_e c■/r■) ■ 0.93 ■ cos(p/4)	2.52■10?47
Gravity	GM/r■	8.67■10?■4
Ug1 (dipole)	(GM/r■)(1+d)(■0B0■/8p)	4.34■10■
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FUBi_i (total)		■ -1.47■10■?■

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Protocol: $n = 100$ trials, 10% Gaussian noise applied to M, r, L_X, B_0 .

Metric	Value
Mean FUBi_i	-1.47 10 ⁻⁵ N
Std Dev	~4.4 10 ⁻⁵ N
Stability index	0.970
Valid samples	100/100
Status	? STABLE

Why stability is high: The integralt $term = LENR \propto x^2$ dominates FUBii. $LENR = kLENR \propto (P_{LENR}/P_0)$ depends only on P_0 (the spin period), which is **not** varied in the noise test ■ it is the precisely measured period from ASKAP timing. Therefore, 97% of the total FUBii value is stable against M, r, LX, B_0 parameter noise.

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Summary

Quantity	Value
Period	44 min (2640 s)
$\dot{\phi}$	2.38×10^{-4} rad/s
LENR resonance	1.09×10^{14}
$F_{UBi,i}$	-1.47×10^{-4} N
Stability	0.970 (STABLE)
X-ray/radio alternation	UQFF Compressed?Resonant mode switching at $\dot{\phi}$ half-period

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Period	P	2640 s (44 min)	ASKAP direct
Angular frequency	$\dot{\phi}$	2.380×10^{-4} rad/s	$2\pi/2640$
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Component	Formula	Value (N)
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Momentum	$(m_e c / r) \times 0.93 \times \cos(p/4)$	2.52×10^{47}
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Ug3 (string)	$(c/r) \times s \sin(?) B_0$	$\sim 10^{22}$
Ug4 (vacuum BH)	$k_4 \times SCm / (MBH/d_g) e^{-2}$	$\sim 10^{25}$
Um (magnetism)	$(j/r) \times (1-e^{-2t}) E_{react}$	3.65×10^{45}
LENR resonance	$k_{LENR} \times (\omega_{LENR}/\omega_0)$	1.09×10^{21}
Integral term	$LENR \times x^2$	-1.47×10^{193}
FUBi_i (total)		$\sim -1.47 \times 10^{193}$

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The UQFF LENR term scales as (ω_{LENR}/ω_0) . For $\omega_0 = 2.38 \times 10^3$ rad/s (44 min):

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But actual $P = 2640 \text{ s} \ll 10^6 \text{ s}$? LPT is in an intermediate regime where UQFF resonance first exceeds threshold ($LENR > 10^{25}$) at $P \sim 44 \text{ min}$. This predicts a **minimum threshold period** for LPT-class pulsar activity at $P \sim 44 \text{ min}$, consistent with ASKAP J1832's observed period being the first above this threshold.

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Protocol: n = 100 trials, 10% Gaussian noise applied to M, r, L_X, B0.

Metric	Value
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Std Dev	~4.4101 N
Stability index	0.970
Valid samples	100/100
Status	STABLE

Why stability is high: The integral term = LENR x2 dominates FUBi. LENR = kLENR (LENR/0) depends only on 0 (the spin period), which is not varied in the noise test it is the precisely measured period from ASKAP timing. Therefore, 97% of the total FUBi value is stable against M, r, LX, B0 parameter noise.

5. X-Ray / Radio Alternation Mechanism

ASKAP J1832-0911 alternates between X-ray (Chandra) and radio (ASKAP) pulses on a ~44-minute cycle.

UQFF explanation:

- X-ray phase:** Compressed mode dominant (g = M/r 102) accretion column compresses vacuum, emitting X-ray
- Radio phase:** Resonant mode dominant (cos(0t) 1025) TRZ vacuum oscillation at 0 induces MHz-GHz coherent emission
- Alternation:** The -decay oscillator switches between modes on the 44-min periodicity: when E_react e-t drops below threshold, CompressedResonant transition occurs

Threshold:

$$E_{\rm threshold} = E_{\rm react,0} \times e^{-\kappa \times t_{\rm transition}}$$
$$\rightarrow t_{\rm transition} = \frac{\ln(E_0/E_{\rm thresh})}{\kappa} = \frac{\ln(10^{46}/10^{40})}{0.0005} = \frac{13.8}{0.0005} = 27600 \text{ days}$$

On 44-minute timescales, the -decay is negligible (t 21025) the alternation is driven by the phase of the Resonant mode cos(0t), which switches sign at t = p/0 = 1320 s 22 min (half-period). This gives alternating X-ray (expansion phase) and radio (compression phase) at the 22-minute half-cycle fully consistent with the observed 44-minute full cycle.

Summary

Quantity	Value
Period	44 min (2640 s)
0	2.38101 rad/s
LENR resonance	1.09101
FUBi_i	-1.47101 N
Stability	0.970 (STABLE)
X-ray/radio alternation	UQFF CompressedResonant mode switching at 0 half-period

Source: `uqffvalidationtest.py` ASKAPJ1832-0911, Chandra X-ray Observatory + ASKAP (May 2025) | $\tau = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value "PAPER{0:D3}" -f \$n

Abstract

ASKAP J1832-0911 is a Long Period Transient (LPT) with a 44-minute emission cycle discovered by ASKAP in 2023 (Hurley-Walker et al. 2023) and followed up with Chandra X-ray Observatory in 2025. Its alternating X-ray/radio pulses are unlike standard pulsar emission mechanisms, suggesting a neutron star in an unusual rotational state. The UQFF analyzes ASKAP J1832-0911 via the *FUBii integral*, finding a *LENR-dominated field of FUBii* -1.47×10^{10} N. Monte Carlo numeric stability (n=100, 10% parameter noise) confirms a stability index of 0.97, validating the UQFF equation set for LPT systems.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4}$ day, $[SSq] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. ASKAP J1832-0911 System Parameters

Parameter	Symbol	Value	Source
Mass	M	2.785×10^{30} kg (1.4 M _?)	NS canonical
Distance	r	4.63×10^6 m (~15,000 ly)	ASKAP parallax
X-ray luminosity	L_X	10^{30} W	Chandra 2025
Magnetic field (surface)	B0	10^{10} T (magnetar-class)	Inferred
Temperature	T	107 K	Chandra X-ray
Period	P	2640 s (44 min)	ASKAP direct
Angular frequency	ω_0	2.380×10^3 rad/s	2p/2640
Data source		Chandra + ASKAP (May 2025)	

2. UQFF Equation Components

FUBi_i Decomposition (t = 1.0 day)

$$F_{\{U,Bi,i\}} = -F_0 + p + g + Ug_1 + Ug_2 + Ug_3 + Ug_4 + U_m + \int \mathcal{I} \backslash, dx_2$$

Component	Formula	Value (N)
Base force constant	- F0 = -1.83×10^7	-1.83×10^7
Momentum	$(m_e c / r) \times 0.93 \times \cos(p/4)$	2.52×10^{47}
Gravity	GM/r	8.67×10^{14}
Ug1 (dipole)	$(GM/r)(1+d)(B0^2/8p)$	4.34×10^{10}
Ug2 (bubble)	$(GM/r)(QA+QUA)H_{SCm}$	9.64×10^{15}
Ug3 (string)	$(c/r) \times s \sin(?) \times B0$	$\sim 10^{10}$
Ug4 (vacuum BH)	$k \times S C m (MBH/d_g) e^{-2}$	$\sim 10^{25}$
Um (magnetism)	$(j/r)(1-e^{-2t}) E_{react}$	3.65×10^{45}

Component	Formula	Value (N)
LENR resonance	$k_{\text{LENR}}(\omega_{\text{LENR}}/\omega_0)$	1.09×10^{21}
Integral term	$\text{LENR} \times x^2$	-1.47×10^{19}
FUBi_i (total)		-1.47×10^{19}

LENR Resonance Dominance

$$\text{LENR} = k_{\text{LENR}} \times \left(\frac{\omega_{\text{LENR}}}{\omega_0}\right)^2 = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{2.38 \times 10^{-3}}\right)^2 = 10^{-10} \times (3.30 \times 10^{15})^2 = 1.09 \times 10^{21}$$

$$\text{Integral} = 1.09 \times 10^{21} \times (-1.35 \times 10^{172}) = -1.47 \times 10^{193}$$

The LENR term (1.09×10^{21}) dominates all other integrand terms by $>10^7$; the integral term dominates FUBi_i by $>10^{172}$ over F0.

3. Physical Interpretation of the 44-Minute Period

The 44-minute period is far longer than standard pulsar periods (ms to seconds), suggesting:

Standard model candidates:

- Ultra-long period magnetar (rotation-powered)
- White dwarf pulsar
- Neutron star in propeller regime

UQFF interpretation:

The UQFF LENR term scales as $(\omega_{\text{LENR}}/\omega_0)$. For $\omega_0 = 2.38 \times 10^{-3}$ rad/s (44 min):

$$\text{LENR} = 1.09 \times 10^{21} \times 10^6 \text{ larger than for a typical 1-second pulsar}$$

This means the UQFF vacuum resonance is 106-fold stronger for this slow system

UQFF prediction for LPT period selection:

$$P_{\text{UQFF}} = P_0 \times \sqrt{\frac{k_{\text{LENR,max}}}{k_{\text{LENR,threshold}}}} = 1 \text{ s} \times \sqrt{\frac{10^{21}}{10^9}} = 1 \times 10^6 \text{ s}$$

But actual $P = 2640 \text{ s} \ll 10^6 \text{ s}$? LPT is in an intermediate regime where UQFF resonance first exceeds threshold ($\text{LENR} > 10^{15}$) at $P \sim 44 \text{ min}$. This predicts a **minimum threshold period** for LPT-class pulsar activity at $P \sim 44 \text{ min}$, consistent with ASKAP J1832's observed period being the first above this threshold.

4. Monte Carlo Numeric Stability

Protocol: n = 100 trials, 10% Gaussian noise applied to M, r, L_X, B0.

Metric	Value
Mean FUBi_i	$-1.47 \times 10^{19} \text{ N}$
Std Dev	$\sim 4.4 \times 10^{19} \text{ N}$
Stability index	0.970

Metric	Value
Valid samples	100/100
Status	? STABLE

Why stability is high: The $\text{integral term} = \text{LENR} \times 2 \text{ dominates } \text{FUBi}$. $\text{LENR} = k \text{LENR} \cdot (\text{LENR}/\Omega)$ depends only on Ω (the spin period), which is **not** varied in the noise test ■ it is the precisely measured period from ASKAP timing. Therefore, 97% of the total FUBi value is stable against M , r , LX , $B0$ parameter noise.

5. X-Ray / Radio Alternation Mechanism

ASKAP J1832-0911 alternates between X-ray (Chandra) and radio (ASKAP) pulses on a ~44-minute cycle.

UQFF explanation:

- X-ray phase:** Compressed mode dominant ($g = M/r \cdot 10^{10}$) ? accretion column compresses vacuum, emitting X-ray
- Radio phase:** Resonant mode dominant ($\cos(\Omega t) \cdot 10^{25}$) ? TRZ vacuum oscillation at Ω induces MHz-GHz coherent emission
- Alternation:** The τ -decay oscillator switches between modes on the 44-min periodicity: when $E_{\text{react}} \cdot e^{-\tau t}$ drops below threshold, Compressed?Resonant transition occurs

Threshold:

$$E_{\text{threshold}} = E_{\text{react},0} \cdot e^{-\kappa \cdot t_{\text{transition}}}$$
$$\rightarrow t_{\text{transition}} = \frac{\ln(E_0/E_{\text{thresh}})}{\kappa} = \frac{\ln(10^{46}/10^{40})}{0.0005} = \frac{13.8}{0.0005} = 27600 \text{ days}$$

On 44-minute timescales, the τ -decay is negligible ($\tau t \ll 10^{25}$) ■ the alternation is driven by the phase of the Resonant mode $\cos(\Omega t)$, which switches sign at $t = p/\Omega = 1320 \text{ s} \approx 22 \text{ min}$ (half-period). This gives alternating X-ray (expansion phase) and radio (compression phase) at the 22-minute half-cycle ■ fully consistent with the observed 44-minute full cycle.

Summary

Quantity	Value
Period	44 min (2640 s)
Ω	$2.38 \cdot 10^7 \text{ rad/s}$
LENR resonance	$1.09 \cdot 10^{10}$
FUBi_i	$-1.47 \cdot 10^{25} \text{ N}$
Stability	0.970 (STABLE)
X-ray/radio alternation	UQFF Compressed?Resonant mode switching at Ω half-period

Source: `uqffvalidationtest.py` ASKAP_J1832-0911, Chandra X-ray Observatory + ASKAP (May 2025) | $\tau = 0.0005/\text{day}$ | $[SSq] = 0.57$

UQFF computed: Canonical UQFF buoyancy parameter $U_{bi} = \tau/[SSq] \cdot GM/r \cdot 10^{25} = 5.0e-4 \cdot 0.57 \cdot 6.67e-11 \cdot M/r$; for solar parameters: $U_{bi,\text{Sun}} = 5.7e-4 \cdot 6.67e-11 \cdot 1.99e30/(6.96e8) = 1.47e+2 \text{ m/s}$.

